

Connect Four Supervised AI Project Report

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• Introduction

The objective of this project is to develop ML models that can play Connect Four. In order to do so, different supervised learning models are given several Connect Four gamestates from the dataset and are tasked with determining whether the gamestate is a winning position (i.e. the player can win regardless of the opponent's next moves provided the player plays perfectly) or not. In theory, by showing one of these models a series of gamestates representing the results of each possible move on a given turn, the model will be able to determine which move(s) are best. Therefore, by repeating this for each turn in a Connect Four game, these supervised learning models will be able to effectively play Connect Four autonomously.

To explore the different approaches to an eventual evaluation, these 5 models were used.

- Logistic Regression
- Support Vector Model (SVM)
- K-Nearest Neighbours
- Neural Network (simple & complex)

The Connect Four Wikipedia article, which describes the game and its rules, can be found [here](#).

• Dataset description

The dataset used in this project is the Connect Four dataset from the UCI Machine Learning Repository. It contains 67,557 examples of valid board positions taken from real Connect Four game sequences.

Key Characteristics

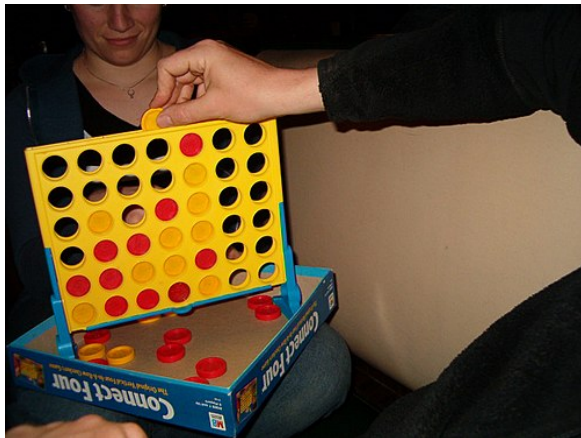
- Instances: 67,557 examples
- Features: 42 features (one for each cell in a 6×7 board)
- Feature types: Categorical — each cell is labeled as:
 - x → first player's piece
 - o → second player's piece
 - b → blank

- Target / Class label:
 - win → first player will win from this position (assuming perfect play)
 - loss → first player will lose
 - draw → game ends tied

Example instance:

```
[b, b, b, b, x, b, b,
b, b, b, b, o, b, b,
b, o, b, x, o, b, b,
b, x, x, o, o, b, b,
x, x, o, o, x, b, b,
o, x, x, x, o, b, b]
```

The above data represents this gamestate:



• Baseline approach description

Here is how each model will behave for the Connect Four problem:

Logistic Regression (LogReg): A linear model that uses a sigmoid function to predict the probability of a gamestate.

Support Vector Machine (SVM): Finds an optimal hyperplane that maximally separates winning and non-winning gamestates in feature space.

K-Nearest Neighbours: Classifies a gamestate based on the majority class among its K most similar gamestates in the training data, where K is an odd integer.

Simple Neural Network (SNN): A shallow multi-layer perceptron with one or two hidden layers that learns non-linear patterns in gamestate features.

Complex Neural Network (CNN): A deeper architecture with multiple hidden layers and more neurons, capable of learning more sophisticated representations of winning patterns.

Below are each of the model's implementations and their tested hyperparameters:

- Logistic Regression
 - This method trains a Logistic Regression model on one-hot encoded Connect Four board states to predict win, loss, or draw. It tests different penalties and C values to find the best model, then evaluates its accuracy, precision, recall, and F1 on the final test set.
 - Train-test split: (60 / 20 / 20)
 - C_values = [0.0001, 0.001, 0.01, 0.1, 1, 10, 100, 1000, 10000, 100000]
 - penalties = ["l1", "l2"]
- Support Vector Machine (SVM)
 - An SVM-based machine learning model that uses one-hot encoded Connect Four board states to classify outcomes by tuning C, kernel, and gamma for best performance.
 - Train-test split: (60 / 20 / 20)
 - C = [0.1, 1]
- K-Nearest Neighbours (KNN)
 - This method uses k-Nearest Neighbours to classify Connect Four board states after one-hot encoding the features. It tests different train/validation/test splits and multiple k values to compare performance across accuracy, precision, recall, and F1.
 - k_values = [3, 7, 13]
- Simple Neural Network
 - This method uses a simple neural network to predict whether position results will be a win, loss, or a draw
 - alpha_values = [0.002, 0.0001, 0.00005]
 - learning_rates = [0.0001, 0.001, 0.01, 0.1]
- Complex Neural Network

- This method trains a multi-layer neural network on one-hot encoded Connect Four board states to predict whether the position results in a win, loss, or draw. It evaluates performance across different train/test splits while measuring accuracy, recall, and precision using a softmax output layer.
- Train-test splits: (60 / 20 / 20), (50, 25, 25), (20, 20, 60)

Competition framework

This competitive framework program loads five trained Connect Four models (two neural networks, KNN, SVM, and Logistic Regression) and simulates games between each pair. It maintains the game board, generates all possible moves, predicts the winning probabilities for each move using the models, and selects the move with the highest chance of winning. After each game, it checks for a winner, updates win counts for each model, and prints the board and cumulative results.

• Evaluation

Logistic Regression

Best Parameters:

- Penalty: 11
- C: 100

Accuracy: 0.7595

	Precision	Recall	F1-Score
Draw	0.6620	0.6334	0.6474
Loss	0.7000	0.005464	0.01084
Win	0.7898	0.9154	0.8480

Support Vector Machine (SVM)

	60/20/20	50/25/25	20/20/60
Accuracy	0.8179	0.8179	0.8062
Precision	0.7986	0.7986	0.7837
Recall	0.8151	0.8151	0.8062
F1 Score	0.7855	0.7855	0.7693

13-Nearest Neighbours

Accuracy: 0.8028

	Precision	Recall	F1-Score
Draw	0.5148	0.1078	0.1782
Loss	0.7846	0.6570	0.7152
Win	0.8151	0.9582	0.8809

Simple Neural Network

Best Parameters:

- Alpha: 0.002
- Learning Rate: 0.01

	60/20/20
Accuracy	0.7573
Precision	0.7089
Recall	0.7573
F1 Score	0.7146

Complex Neural Network

	60/20/20	50/25/25	20/20/60
Accuracy	0.8380	0.8238	0.8076
Recall	0.8147	0.7996	0.7819
Precision	0.8066	0.8485	0.8347
F1 Score	0.8369	0.8233	0.8075

Overall model performance comparison at the 60-20-20 train-test split

	Accuracy	Precision	Recall	F1 Score
Complex Neural Network	0.8380	0.8066	0.8147	0.8369.
Simple Neural Network	0.7573	0.7089	0.7573	0.7146
K-Nearest Neighbours	0.8028	0.8151	0.9582	0.8809
Support Vector Machine	0.8179	0.7986	0.8151	0.7855
Logistic Regression	0.7595	0.7898	0.9154	0.8480

Model competition (# wins)

	Original	P2 tries to choose X's worst move	Flipped board	LogReg on confidence
Complex Neural Network	5	6	4	4
Simple Neural Network	1	2	2	2
K-Nearest Neighbours	3	4	4	3
Support Vector	6	2	8	8

Machine				
Logistic Regression	5	6	2	3

• Discussion

Among the selected models at the 60/20/20 train-test split, the complex neural network had the highest accuracy, followed by the SVM, then the 13-nearest neighbours, then the logistic regression model followed very closely by the simple neural network. However, the models did not deviate hugely in accuracy, ranging from 0.8380 (CNN) to 0.7573 (SNN). Precision among all models was very close, roughly between 0.79 and 0.82, except for the SNN, which had a comparatively low precision of 0.7089. Recall varied much more, with the SNN having 0.7573 recall, then the SVM and CNN each very close to 0.815, and finally the LogReg at a rather high 0.9154 and the 13-nearest neighbours at a staggering 0.9582. Overall, the CNN, SVM, and 13-nearest neighbours were very performant, while the SNN and LogReg model were notably less performant.

These statistics deviate slightly from the model competition results. Expectedly, the SVM and CNN performed very well, with the SVM netting the most wins at six won games and the CNN closely following at five games. Meanwhile, the SNN won only one game, similarly matching its rather poor performance. However, the 13-nearest neighbours performed much more poorly than the CNN and SVM at three won games, while the LogReg model excelled, tying with the CNN at five won games. When player 2 was attempting to choose player 1's worst move, the SVM performed notably worse, tied with the SNN for the lowest performance at 2 wins each. The CNN and LogReg were tied for the best performance at 6 wins each, while 13-nearest neighbours sat at the mean of 4 wins. With a flipped board, though, the SVM performed exceptionally at 8 wins, with 13-nearest neighbours and the CNN following from a distance at 4 wins each, and the SNN and, surprisingly, the LogReg model at a very low 2 wins each.

A very rough correlation between the test set performance metrics and the competition results can be seen, with the SVM and CNN alike performing well in both competition and in test set performance, while the SNN performed poorly in both, as well. However, inversely, the 13-nearest neighbours had rather mediocre performance in the competition despite its performance in the test set, while the logistic regression model performed very well in the competition despite its much poorer test set performance, performing worse than the 13-nearest neighbours in every single performance metric despite netting more wins in both the original

competition and as player 2 choosing player 1's worst move. This may indicate a poor correlation between test set performance metrics and competition results, but this is hard to say, given that the deviation across all four performance metrics in the test set (accuracy, precision, recall, F1 score) was not very high, and given that the number of tested models was just five.

• Conclusion

In the testing sets, the CNN achieved the highest accuracy, followed closely by the SVM and KNN models. Accuracy roughly ranged from 75% to 84% whereas precision hovered around 80%, except for the SNN. Recall values vary widely from 75% (SNN) to 95% (CNN). Generally, the SVM and CNN were the most performant, whereas the 13-nearest neighbours slightly trailed and the logistic regression and especially the SNN were the least performant. For competitive performance, the CNN and SVM showed the greatest performance among all tests, while the logistic regression model showed similar performance aside from the flipped board competition. The 13-nearest neighbours and SNN consistently performed at a mediocre and poor level, respectively. Testing performance roughly aligned with competition results, with the CNN and SVM excelling, the SNN underperforming, and the 13-nearest neighbours somewhere between. However, the slight discrepancy in the 13-nearest neighbours' testing and competition performance and the major discrepancy between the logistic regression model's testing and competition performance highlight that adversarial performance can differ greatly from test metrics.

• Appendix: contribution of each team member

Blake Doan:

- Co-found project dataset
- Developed SVM model
- Co-developed model competition framework

Sam Kramer:

- Co-found project dataset
- Developed Complex Neural Network
- Co-developed model competition framework

Keiron Coolen:

- Developed K-Nearest Neighbours model
- Co-wrote project report

Malec Tarabein:

- Developed Simple Neural Network
- Co-wrote project report

Juan Javier Denoso:

- Developed Logistic Regression Model
- Designed presentation slideshow